

1. Business Context

THEHOTEL.com operates 38 hotel brands and receives tens of thousands of guest reviews each month. Management cannot act on individual reviews at scale. The business need is to identify which operational areas drive dissatisfaction so the operations team can prioritise remediation.

Business framing. Three questions guide this analysis: (1) which operational areas generate the most dissatisfaction across brands, (2) whether dissatisfaction patterns differ by guest segment, and (3) whether complaint themes are shifting over time.

Business value. Identifying the top complaint areas allows the group to distinguish issues requiring group-wide service standards from brand-specific property problems, and to direct resources accordingly rather than applying generic improvement programmes.

2. Dataset and Cleaning

The dataset contains 47,741 guest reviews across 38 brands, collected between May and June 2023. Reviews are bilingual: 60.3% English, 39.7% French. Each row includes review text, date, hotel code, brand code, language, trip type, gender, and age group.

Cleaning steps: whitespace stripping, normalisation of synonymous categorical values, and deduplication on hotel code, date, and text. Missing demographics (gender: 3.0%, age: 5.3%) were kept as **not disclosed** rather than imputed, since both are self-reported. No missing text or trip type was found.

3. Exploratory Data Analysis

Review volume is stable across the two months with no strong weekly seasonality. Couple trips are the most common segment (38.1%), followed by family (24.6%) and business (19.8%). The 35–54 age group accounts for 41.3% of reviews. Language split varies by brand: five brands have a French review share above 80%, suggesting distinct regional guest audiences that warrant separate consideration in downstream analysis.

4. Sentiment Analysis

4.1 Model selection

Two models are used. XLM-RoBERTa [1] is the primary model: it handles English and French natively without translation, which is essential for this bilingual dataset. VADER [2] is applied to English-only reviews as a secondary comparison, since its lexicon is English-only and produces uninformative near-neutral scores on French text.

4.2 Results

XLM-RoBERTa classified 68.9% of reviews as positive, 29.2% as negative, and 1.9% as neutral. Agreement with VADER on English reviews is 78.8%, with VADER over-classifying reviews as pos-

¹Code available at <https://github.com/wengchienwei/thehotel-voc-analysis>

itive (24,290 vs. 19,236 by XLM-RoBERTa). XLM-RoBERTa is used for all downstream analysis. The overall negative share dropped from approximately 40% in early May to 28% by mid-May and stabilised through June (Figure 1). Segment analysis (Figure 2) shows that business travellers have the highest negative rate at 35.5%, approximately 10 percentage points above single travellers. The 25–34 age group reports a 34% negative rate, the highest of any age group.

5. Negative Review Theme Extraction

5.1 Method

BERTopic [3] is applied to the 8,041 high-confidence negative reviews (XLM-RoBERTa score ≥ 0.745 , i.e. above the mean confidence threshold). The threshold removes ambiguous borderline negatives that cluster poorly. Embeddings use `paraphrase-multilingual-MiniLM-L12-v2` [4], which handles English and French natively. A combined English–French stop-word list is passed to the `CountVectorizer` to prevent stop words from dominating topic labels.

5.2 Topics and trend

Three topics emerge:

- **room_and_service:** Room condition, cleanliness, housekeeping, and staff responsiveness. 55.9% of all negative reviews.
- **ibis_value_ratio:** Room size, dated facilities, and price-quality mismatch relative to Ibis Budget. 1.6% of all negative reviews, concentrated in brands 2 and 0.
- **food_and_breakfast:** Breakfast quality and food service. 0.2% of all negative reviews, concentrated in brand 19 and dominated by French-language reviews.

The `room_and_service` share held stable at approximately 98% of weekly negative reviews across the full two months with no improvement, confirming that complaint themes are structural rather than transient. Brand concentration is shown in Figure 3.

6. Business Recommendations

1. **Group-wide: set minimum service standards for business traveller segments.** Business travellers have the highest negative rate (35.5%) and complaints concentrate on room readiness and staff responsiveness. This is the highest-impact, most implementable action.
2. **Group-wide: investigate room_and_service complaints across brands 35, 28, 33, and 31.** These brands show 20–28% of their total reviews flagged in this topic. A shared root cause analysis is more efficient than four separate brand audits.
3. **Brand-specific: audit brand 36.** Brand 36 has a 64% negative rate on only 25 total reviews, with 75% of its negatives on room and service. A targeted property review is appropriate rather than a group programme.
4. **Brand-specific: review room quality and pricing for Ibis brands 2 and 0.** Guests compare these properties unfavourably to Ibis Budget at similar price points. Actionable through room refurbishment and clearer product tier communication.
5. **Brand-specific (low priority): brand 19 breakfast offering.** French-speaking guests flag breakfast quality. Low volume (0.2% of all negatives); not an urgent group-wide action.

References

- [1] Cardiff NLP. *XLM-RoBERTa Base Sentiment*. Hugging Face Model Card.
<https://huggingface.co/cardiffnlp/twitter-xlm-roberta-base-sentiment>
- [2] NLTK Project. *VADER Sentiment Analysis*. NLTK Documentation.
<https://www.nltk.org/api/nltk.sentiment.vader.html>
- [3] Grootendorst, M. *BERTopic*. BERTopic Documentation.
<https://bertopic.com/>
- [4] Sentence Transformers. *paraphrase-multilingual-MiniLM-L12-v2*. Hugging Face Model Card.
<https://huggingface.co/sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2>

Appendix: Figures

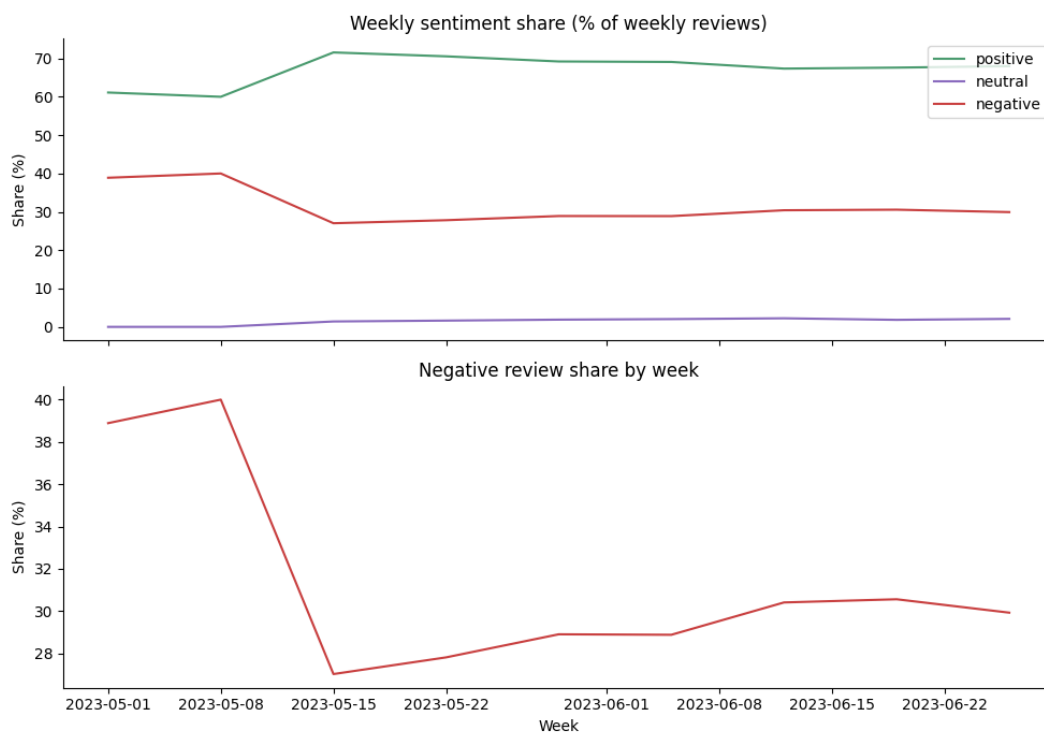


Figure 1: Weekly sentiment share and negative review share. Negative share dropped from $\sim 40\%$ in early May to $\sim 28\%$ by mid-May and stabilised.

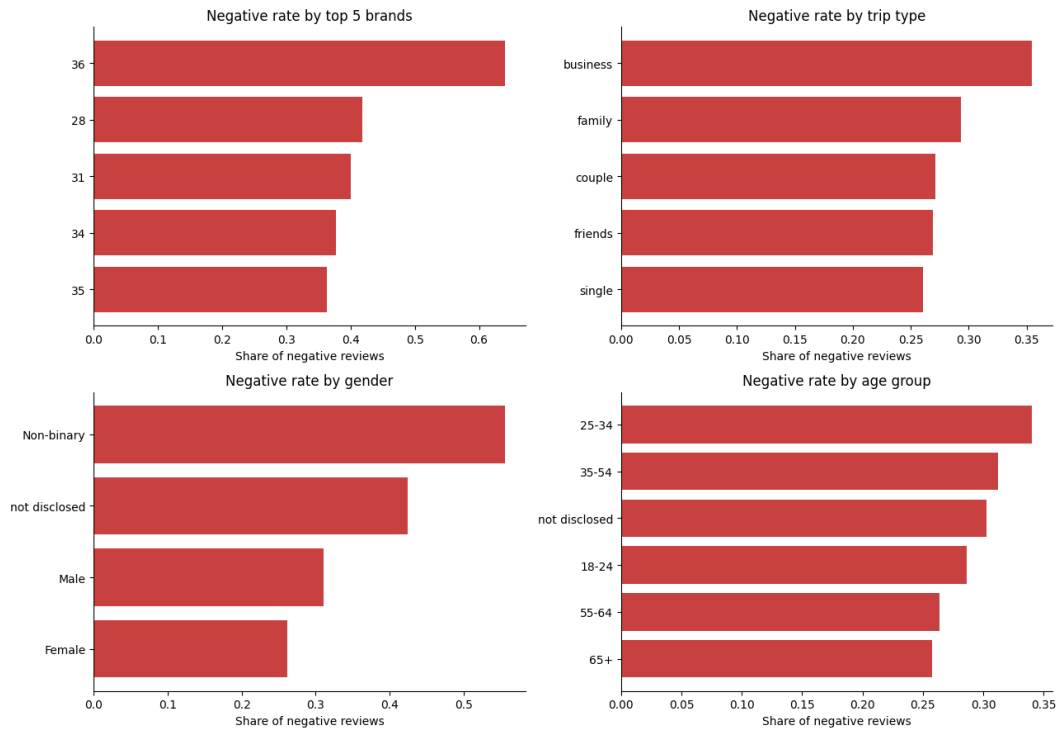


Figure 2: Negative rate by brand, trip type, gender, and age group. Business travellers and the 25–34 group are the most dissatisfied segments. Brand 36 is the outlier at 64%.

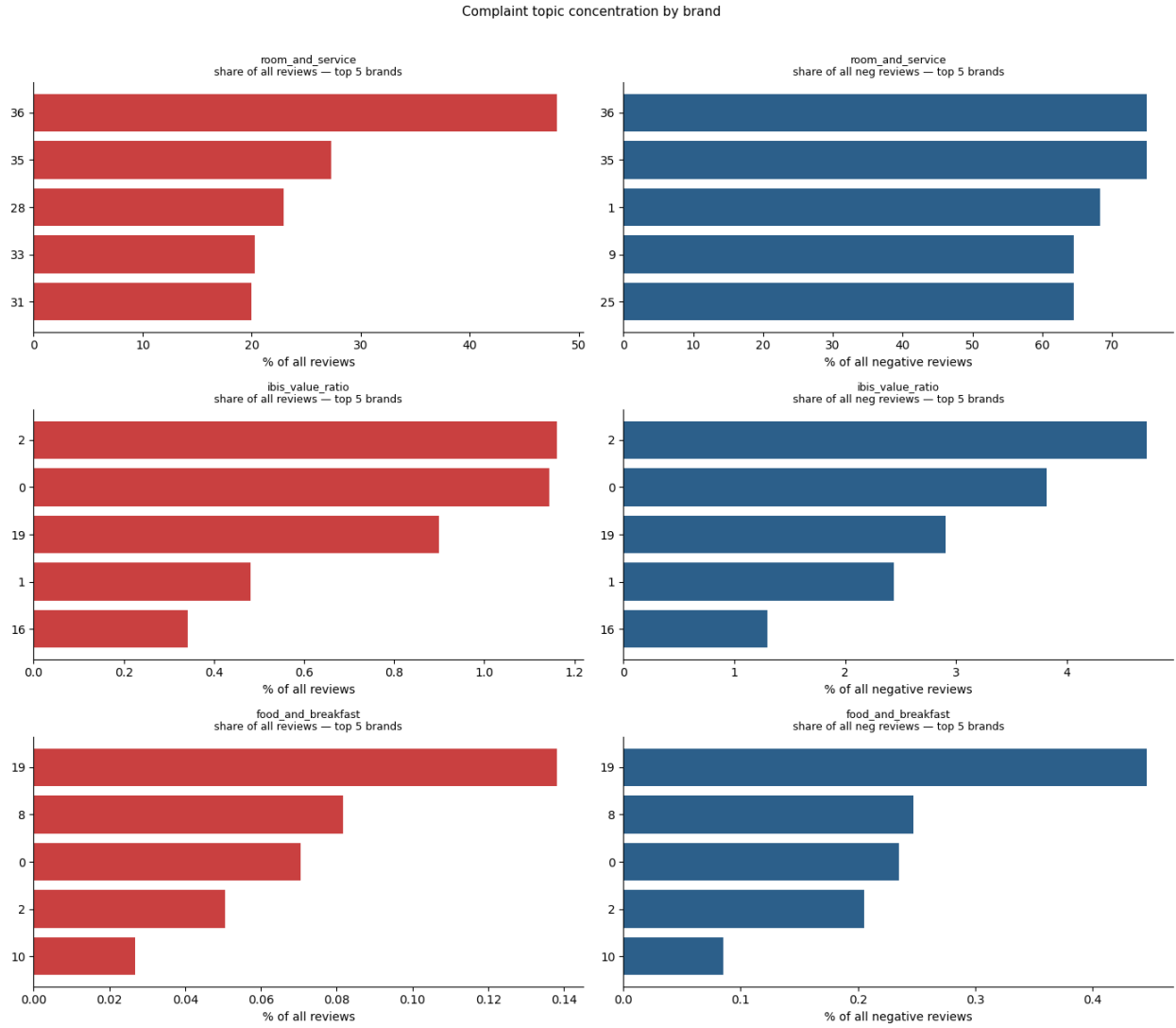


Figure 3: Complaint topic concentration by brand. Left: share of all reviews. Right: share of all negative reviews.